

LEWISSETTA, VA, USA

WMO: 997304

Lat: 37.9954N

Lon: 76.4646W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.2	(c) -4.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 12.7	(k) 4.5	(l) 11.9	(m) 2.8	(n) 5.3	(o) 350	(p) 0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.2	(c) 32.3	(d) N/A	(e) 31.1	(f) N/A	(g) 30.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.2	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.0	(b) 10.4	(c) 8.8	(e) -8.9	(f) 34.8	(g) 2.7	(h) 1.1	(i) -10.8	(j) 35.6	(k) -12.4	(l) 36.3	(m) -13.9	(n) 37.0	(o) -15.8	(p) 37.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	3.4	4.4	8.1	14.5	19.4	24.1	26.4	25.5	22.3	16.7	10.2	6.5
	DBStd	9.01	5.36	5.01	4.90	4.35	3.72	2.82	2.09	1.98	2.80	3.95	4.07	4.70
	HDD10.0	671	215	169	98	9	0	0	0	0	0	3	46	131
	HDD18.3	2027	463	391	320	131	33	1	0	0	3	77	243	367
	CDD10.0	2562	11	12	38	143	292	424	507	481	368	211	53	22
	CDD18.3	878	0	1	1	16	66	175	249	222	121	27	1	1
	CDH23.3	6348	0	0	5	88	322	1225	2279	1761	591	78	0	0
	CDH26.7	1681	0	0	0	17	62	324	723	439	109	7	0	0
Wind	WSAvg	4.2	4.4	4.5	4.7	4.6	4.2	3.8	3.8	3.7	3.9	4.3	4.2	4.1
Precipitation	PrecAvg	1176	84	74	101	85	94	102	132	133	118	93	72	89
	PrecMax	1674	174	197	230	155	181	206	232	296	408	191	229	206
	PrecMin	890	32	22	8	36	24	43	35	43	15	3	4	32
	PrecStd	177	33	39	41	28	39	36	48	66	86	47	49	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	21.0	24.0	29.0	30.7	33.5	34.7	33.0	31.5	27.4	22.1	21.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	17.2	17.9	20.7	25.8	28.5	31.1	32.9	31.5	29.4	25.2	19.8	19.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	14.8	15.0	18.3	23.4	26.6	29.9	31.5	30.3	27.8	23.9	18.0	16.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	11.9	11.7	15.7	21.2	25.1	28.6	30.3	29.1	26.5	22.6	16.5	13.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	7.1	7.4	6.7	6.2	6.2	5.6	5.4	6.0	6.2	6.2
		MCDBR	10.8	11.3	11.7	10.5	8.5	7.9	7.9	7.3	7.1	7.2	8.3	9.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.309	0.319	0.352	0.392	0.433	0.483	0.515	0.508	0.410	0.358	0.325	0.312
	taud		2.527	2.490	2.438	2.339	2.262	2.166	2.092	2.097	2.351	2.496	2.545	2.554
	Ebn at Noon		879	913	910	887	853	807	777	772	838	856	852	851
	Edh at Noon		77	91	105	123	136	149	160	155	114	89	76	71
All-Sky Solar Radiation	RadAvg		2.24	3.03	4.05	5.12	5.82	6.36	6.13	5.51	4.46	3.43	2.53	1.98
	RadStd		0.17	0.33	0.34	0.36	0.61	0.46	0.29	0.29	0.43	0.38	0.22	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (76 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+150	+94	

Nomenclature: See separate page